N4 Version 1 & 2 – Final Test Campaign & Launch Preparation

Nakuja Rocketry Club
Solid Propulsion Team
Updated Critical Path Documentation
N4 Version 2 – Final Test Campaign & Launch Preparation

Document Version: 3.0
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1. Campaign Objectives

Priority	Motor Configuration	Casing	Static Test Target Date	Purpose
1	3-grain	Aluminium 6063-T5	**Monday 17 August**	Intermediate verification
2	4-grain	Stainless Steel G304 SCH 10	**Friday 21 August**	High Margin Option
3	5-grain	Aluminium 6063-T5	**Monday 24 August**	Primary flight candidate

Post-success: Mass-produce two flight motors of each successful configuration for dual launch (minimum two rockets).

Hard deadline: All static tests completed and flight motors ready for launch by **11 September 2026**.

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2. Critical Path Overview (from 10 August)

Main Sequence:

1. Materials available (10 Aug)

2. Parallel start: Core rods + Machining (nozzles + bulkheads) + Casting tool printing + Grain cooking
3. 3-grain assembly & static test → **17 Aug**
4. Immediate transition to Stainless Steel 4-grain cooking/assembly → static test **21 Aug**
5. Aluminium 5-grain path (parallel where possible) → static test from **24 Aug week**
6. Mass production of successful configurations

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3. Detailed Timeline

Phase 0 – Materials & Preparation (10–11 August)

- Receive and inventory all purchased items
- 3D-print additional casting tools (target: 3–4 tools total)
- Machine first set of core rods
- Begin liner fabrication (if epoxy is available)

Phase 1 – 3-Grain Aluminium Motor (Target Static Test: 17 August)

Task	Owner	Start	Duration	Notes
Machine core rods + motor preparation (drilling, tapping)	Machining Team	10 Aug	3–4 days	Parallel where possible
Cook 3 grains	Propellant Team	10–12 Aug	1 day each	Max 1 grain/day initially
Cure grains	—	—	7 days	Critical path item
Fabricate liners	Propellant Team	10–12 Aug	1 day each	
Dry assembly	Machining + Propellant	14–15 Aug	1–2 days	
Wet assembly + ignitor	Full team	16 Aug	1 day	

Task	Owner	Start	Duration	Notes
Static Fire – 3 Grain	Full team	**17 Aug**	1 day	Data review same day

Phase 2 – 4-Grain Stainless Steel Motor (Target Static Test: 21 August)

Task	Owner	Start	Duration	Notes
Machine nozzle + bulkhead	Machining Team	18–22 Aug	3–4 days	Parallel with 3-grain work
Cook 4 grains	Propellant Team	18–21 Aug	4 days	
Cure + Assembly	Full team	22–25 Aug	—	
Static Fire – SS 4 Grain	Full team	25–28 Aug	1 day	Flexible within week

Phase 3 – 5-Grain Aluminium Motor (Week of 24 August)

Task	Owner	Start	Duration	Notes
Cook 5 grains	Propellant Team	13–17 Aug	5 days	Overlap with 4-grain cure
Cure grains	—	—	7 days	Must finish by 20 Aug
Assembly (dry + wet)	Full team	19–20 Aug	2 days	
Static Fire – 5 Grain	Full team	**21 Aug**	1 day	Critical decision point

Phase 4 – Mass Production (from 24 August / after successful tests)

- Replicate successful Aluminium 5-grain configuration (priority)
- Replicate successful Stainless Steel 4-grain configuration
- Target: 2 flight-ready motors of each successful type

- Re-use proven casings, nozzles, and bulkheads where possible to compress timeline

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4. Resource & Constraint Notes

- **Grain production rate:** Initially limited to 1 grain/day (1 core rod + 1 casting tool). Additional tools must be printed immediately on 10 August to increase throughput.
- **Curing remains the longest single constraint** (7 days). Overlapping cooking of the next motor's grains while the previous set cures is mandatory.
- **Machining:** Nozzles (3 days), bulkheads (2 days) and core rods (1 day) must be scheduled aggressively in parallel across the two aluminium motors and the stainless-steel motor.
- **Ignitors:** Can be prepared in a single day and done in parallel with grain work.

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5. Decision Gates

Date	Gate	Possible Outcomes
17 Aug	3-grain Aluminium result	Proceed / Hold / Adjust
21 Aug	Stainless Steel 4-grain result	High Margin of error backup
25–28 Aug	5-grain Aluminium result	Use results from 3 grain to produce 5 grain

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6. Team Responsibilities

- **Propellant Team (2):** Grains, liners, ignitors, chemistry
- **Machining Team (2):** Nozzles, bulkheads, core rods, assembly support
- **Ignition Team:** Circuit fabrication, motor ignition, data acquisition
- **3D Printing (1):** Casting tools, any urgent housings
- **Documentation (1):** Daily logs, test reports, GitHub updates

7. Success Criteria for Launch Readiness

- At least one configuration (Al 5-grain or SS 4-grain) successfully static-tested
- Two flight motors of the chosen configuration fully assembled and inspected
- All data packages and safety documentation complete